Rapid Planetary Defense

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System of Systems Grand Challenge 23
Work in Progress

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Rapid Planetary Defense

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Rapid Planetary Defense

Executive Summary

A system of systems (SoS) is defined as a set or arrangement of systems that results when independent and useful systems are integrated into a larger system that delivers unique capabilities[**SoSE-DAG**]. If you don't leave a blank line, LaTeX won't go to the next line, so you can put sentences in different lines and commands in between sentences.

Contents

Nomenclature

Abbreviations

$DART$	Double Asteroid Redirection Test
ESA	European Space Agency
kT	Kilotons
MT	Megatons
$NASA$	National Aeronautics and Space Administration
NEO	Near Earth Object
NEO	Near Earth Orbit
$PDCO$	Planetary Defense Coordination Office
PJ	Peta Joule
TT	Teratons
SoS	System(s) of systems

Symbols

KE	Kinetic Energy
m	Mass
v	Velocity

Chapter 1

Introduction & Motivation

Over the course of history, Earth has been hit countless times by extraterrestrial objects [1]. The severity of those impacts was widespread, ranging from little to no effects in instances when objects vaporized and disintegrated into objects of significantly less mass during entry into Earth's atmosphere. Due to acting friction forces, the air resistance, during the high-velocity entry of objects into the atmosphere, extreme heat is generated, effectively partially melting and vaporizing the object, reducing its weight while simultaneously decelerating it [2]. Depending on the object's composition and size, a significantly smaller object or numerous even smaller objects eventually hit the surface. The severity of actual surface impacts corresponds with an object's impulse at impact. On first thought, a mid-air breakup may seem advantageous since the impulse is directly proportional to an objects mass. However, the in-flight atmospheric disintegration can release an enormous amount of energy, creating a potentially devastating pressure and heat wave, comparable to those of nuclear detonations [3]. The impact frequency of objects that pose a significant thread to the existence of life on Earth is slim, however impact or mid-air disintegration of objects which can cause extensive local and even regional destruction occurs much more regularly [4]. Several countries and space agencies such as the National Aeronautics and Space Administration (NASA) and the European Space Agency (ESA) have already established offices, research groups, to investigate, plan and design possible planetary defense options against extraterrestrial threats [5] [6]. However, the main challenge lays in the generation of a comprehensive system of systems that includes all components required, their (inter-) dependencies and behavior, involved stakeholders and their politics, as well as the consideration of a great number of possible threat scenarios. A point solution technology and mission design does not represent a suitable methodology to comprehensively analyze the combinatorial threat space and the required reaction chain to execute threat mitigation, nor to detect the general need for system and technology development in order to guarantee reliable planetary defense. Furthermore, current efforts remain intra-disciplinary and do not account for external factor such as politics and general stakeholder agendas.

1.1 Historical Background

The perhaps most prominent example of a direct surface impact of an extraterrestrial object, an asteroid, occurred approximately 66 million years ago. The so called K-T impactor, a so called "global killer" that posed an "extinction level event" had a diameter of roughly 9 km and had an impact velocity of 20 km/s [7] [4]. The energy released during that particular event is estimated to be of the order of 100 TT (teratons) of TNT equivalent, or 4.5 billion times the explosive power of the Hiroshima atomic bomb. The direct consequences were devastating, superheated winds propagated at supersonic speeds, destroying all vegetation, killing most life in a radius of up to 1800 km around the impact side. Several trillion tons of molten material were ejected from the impact side and rained down all over the world causing the destruction of 70 % of the world's forests. Since the asteroid impacted in the Golf of Mexico, tsunami waves taller than 1.5 km washed ashore and the impact's shock caused earthquakes and volcanic erupts worldwide. Although the direct "kill-zone" was limited to a radius of several thousand kilometers the most significant effect of the impact, that eventually caused the extinction of 70 % of the species world wide was the so called "nuclear-winter effect". Dust ejected from the impact side and by worldwide volcanic activity, saturated the atmosphere, effectively shielding sunlight, causing a significant and rapid decrease of the global temperature [7] [8]. On a macro timescale, a temperature inversion was the final consequence of the impact. Wildfires and vaporized material that has been ejected from the impact side, caused drastic global warming due to the subsequent greenhouse effect [8].

More recent impact events in Earth's include e.g. an asteroid impact in northern Arizona approximately 50,000 years ago. The impact of an asteroid with an estimated diameter between 30 m and 50 m caused an energy release of 20 to 40 MT (megatons) of TNT equivalent. This impact did not cause effects on a global scale, but most likely severe regional devastation. The impact crater has a diameter of nearly one mile, ejected rocks from the impact side rained down within a radius of up to 2 km, the resulting shockwave propagated with supersonic speed and flattened trees in a 14 km radius [9].

Figure 1.1: Impact Crater in Northern Arizona [10]

An comparable event occurred in 1908 over rural Siberia, Russia, when an asteroid of approximately 40 m in diameter explosively disintegrated at an altitude of 10 km over Tunguska. The energy yield is estimated to have been approximately 15 MT of TNT equivalent. The resulting blast wave flattened the trees across a surface area of 2149.69 square kilometers [11].

Figure 1.2 showcases the number of known airburst events of asteroids with diameters smaller than 20 m from 1908 till 2013. It is apparent that no location on Earth is safe against asteroid events and that, although the Tunguska and the most recent event took place over rural Russia, airbursts have occurred over more densely populated areas over the last century. As for reference, the largest diameter dot represents the most recent airburst, recorded in 2013. The energy released was approximated to be of the order of 1 PJ (peta joule) which translates to 300 kT TNT equivalent, 20 times the yield of the Hiroshima atomic bomb.

The most recent atmospheric entry of an asteroid took place in 2013. An asteroid of 20 m in diameter disintegrated over the Russian city of Chelyabinsk, releasing an approximate

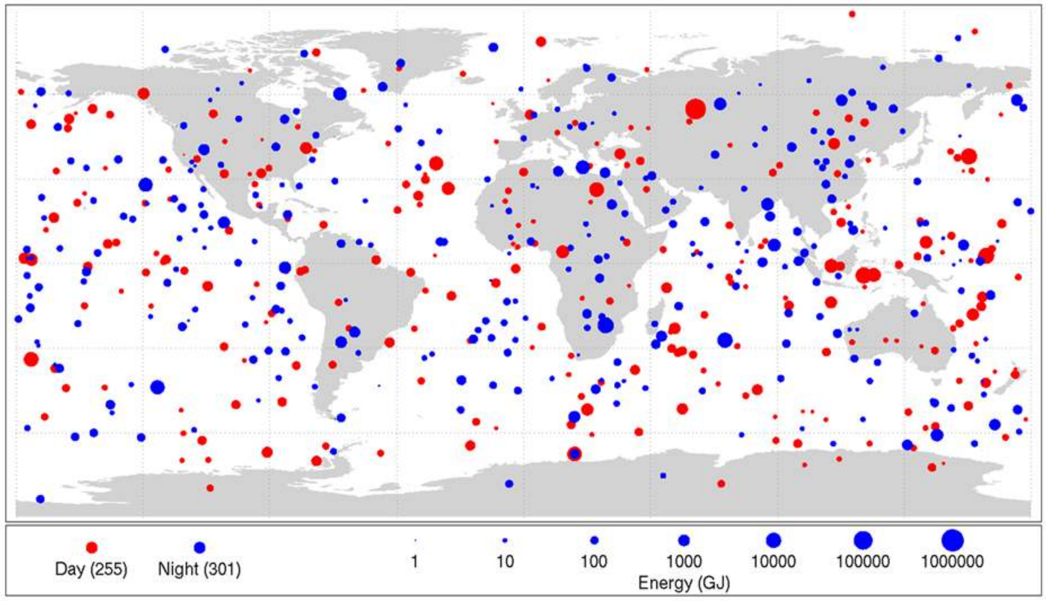

Figure 1.2: Airburst events of Asteroids ¡ 20 m from 1908 till 2013 [4]

energy equivalent of 500 kT TNT equivalent [12]. The resulting airburst and shockwave damaged over 7000 buildings and injured over 1500 people [4]. Although the asteroid has been detected months prior to entry into Earth's atmosphere, no mitigation or evacuation efforts have been initiated, showcasing the current degree of helplessness due to the lack of prepared emergency and planetary defense plans that allow for a timely, flexible and threat-specific execution of counter measures [13].

As direct response to the Chelyabinsk event of 2013, NASA established the Planetary Defense Coordination Office (PDCO) in 2016. The PDCO coordinates global planetary defense efforts and conducts technology studies to analyze and evaluate defense strategies against so-called near earth objects (NEOs) [12]. In 2022 NASA in collaboration with ESA executed the first and so far only test of a mitigation technique.The Double Asteroid Redirection Test (DART) mission, brought forward the proof of concept, that redirection of asteroids through utilization of a kinetic impactor in form of a spacecraft is possible, that the asteroids trajectory can be distinctively changed by human intervention [14].

1.2 Future Implications

Till today, humanity has had luck, that no asteroid impact or mid-air explosive disintegration occurred over densely populated areas. Overlaying the approximate energy equivalent of the

Chelyabinsk event from 2013 and the impact in Arizona from 50,000 years ago and onto a map of New York city showcases the potentially devastating consequence of such events. An equivalent to the 2013 event could devastate New York's city region (left image), whereas an equivalent to the Arizona impact would destroy the whole New York city metro area (right image). This implies a potential severe loss of human life, since New York City has more than 8 million inhabitants, enormous loss of property and an indeterminable impact on the nations economical market [15].

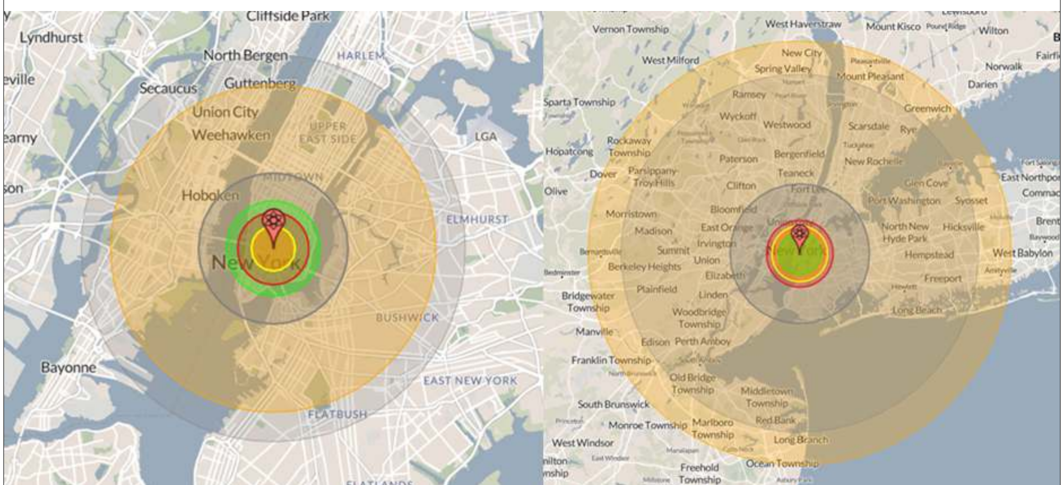

Figure 1.3: Potential Devastation by Impact Events - New York City [16]

Asteroid impacts on Earth are inevitable and only a question of time. Without adequate mitigation strategies, it is not possible to either avoid or actively control the threat of impacts or explosive mid-air disintegrations. Extrapolation, based on historical data, to determine the time of the next event is not possible and the accuracy of entry trajectory propagation of threats that have a high entry probability is not as accurate as required to determine early on if the event affected location is densely populated or not [1]. Additionally, it is apriori not possible to determine weather or not an asteroid will airburst. Even if the most probable entry trajectory is approximated, evacuation of densely populated areas along the trajectory path will most likely be unrealistic due to a limited warning time.

In the future it is unlikely that humanity will face an extinction level asteroid threat. From a statistical standpoint such an event is expected to occur every hundred million years, furthermore due to the size of such objects, it is highly likely that those will be detected a long time in advance [4] [17]. However, detection of the more imminent planetary threats, which are less than several hundred meters in diameter is more challenging. From the population of objects that are considered to be close to earth, near earth objects (NEOs), it is assumed that 95 % of those with a diameter greater than one kilometer, and therefore capable of devastation on global scale, have been detected [1]. But only 40 % of NEOs smaller than 140 meters in diameter, and thus capable of severe regional destruction, have been detected [1]. Asteroids, comparable to the specimen that entered Earth's atmosphere in 2013, are expected to reoccur every 80 years [4].

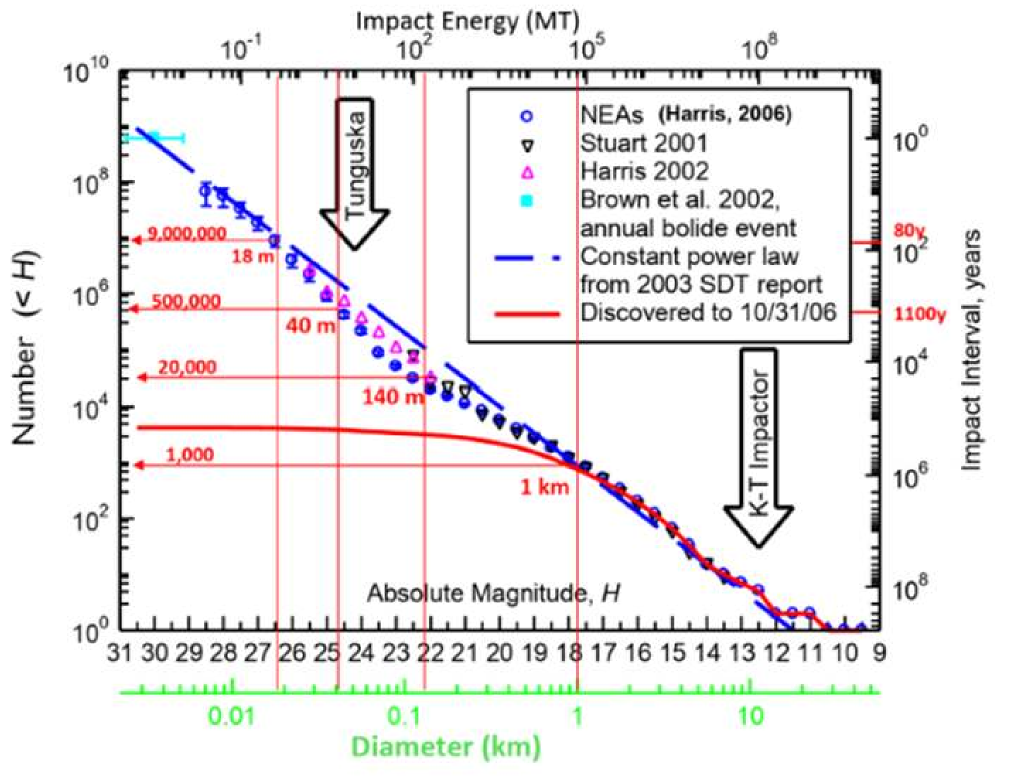

Figure 1.4: Near Earth Asteroid Population, Impact Frequency vs. Impact Magnitude [18]

However, current numbers can cause wrong impressions. Since detection capabilities are inherently limited, the total number of NEOs can only be estimated. For example, objects with elongated orbits may not have been detected because they are currently at their greatest distance to their central body. Additionally, due to astrodynamic instabilities, NEOs that were previously determined to not pose a threat to Earth can become a threat, e.g. through collisions within asteroid belts. Furthermore, when determining threats to earth, another class of objects has to be taken into consideration. Those are interstellar objects which originate from outside the solar system. Due to their interstellar trajectory, early detection is currently limited since orbit observation is selective because of limited observation capabilities that are available. Furthermore, accurate orbit determination and propagation is limited [19].

1.3 State of the Art

Because there are many types of threats, there is no one-size-fits-all mitigation strategy for dealing with NEO objects. There are many moving pieces, and many parties involved, and depending on the threat type different communication structure must be followed. NASA does not have the full scope of this problem, and comprehensive SoS consideration is essential to gain informed knowledge and make informed and subsequently appropriate decisions. NASA's 10-year plan is primarily focused on global communication and mitigation strategies; however, these are more or less just point solutions.

A large issue with the current state-of-the-art is the defunding of agencies and programs related to the development of mitigation and detection strategies. While defunding may occur eventually, an idea of the bare minimum to cover the space around us must be known first. Another question brought up is regarding who is responsible for the defense of other countries, and this is something current government stakeholders are discussing. Is each country only responsible for itself and its allies, or is it also responsible for each island country that does not have its own mitigation capabilities?

There has been one mitigation mission in the past, the DART mission. However, this mission mainly served as a proof of concept, as no more money or work has been poured into this project since its completion. The DART mission was very successful; however, it was capable of changing the orbit of the Didymos asteroid by 32 minutes. Before the mission, scientists set the minimum orbit change to be 73 seconds, and the DART mission completely blew this out of the water, exceeding it by more than 25 times [20]. DART was launched on November 24, 2021, with a Falcon 9 rocket, and it traveled for over 10 months before finally making contact with the Didymos asteroid [21]. It took three years to construct and launch [22].

1.4 Motivation

The motivation of this project is derived from the provided historical background and the future implications that clearly identify, that an entry event of an extraterrestrial body is inevitable and, without intervention, uncontrollable, posing a significant threat to human life, infrastructure, and economy on a regional (national) as well as potentially a global (international) scale. The overarching motivation is therefore, the formulation of a system of systems framework with the top level requirement to protect Earth and thus humanity from extraterrestrial threats. Furthermore, all participating disciplines involved must be accounted for and considered, to establish a comprehensive representation of the required system of systems within the framework. Especially, since time is the most essential resource within the process of planetary threat mitigation and or the evacuation efforts, the framework must enable streamlined communication between all stakeholders involved. Current mitigation strategy development does not provide a comprehensive configuration of all entities and aspects that would be involved in such efforts. This furthermore implies that current strategy development builds upon a partially incomplete system, and derived system requirements, e.g. in terms of detection capabilities and minimum required warning times are inadequate. The overarching motivation and the derived top level requirement is subsequently broken up into different aspects of planetary defense efforts.

1.4.1 Threat Mitigation

The vast combinatorial variety of different extraterrestrial threat types and detection capabilities, hence warning time frames, requires a comprehensive consideration of threat-types and mitigation strategies. Depending on the warning time and threat-type at hand, several mitigation strategies might not be feasible, thus should not be taken into consideration. The applicability of mitigation strategies for a given warning time and threat at hand, should be known apriori in order to streamline and focus available resources only onto feasible mitigation options right from the beginning, after detection.

1.4.2 Political

Since time is most essential for the initiation and execution of planetary threat mitigation mission, means of streamlined communication of plans of action is critical. Decision makers on the national and international level must be able to quickly react and decide to ensure planetary safety. Although planetary defense implies the involvement of every nation, nation-specific political agendas and motivations must be considered respectively. Questions of "what is required to ensure national protection against extraterrestrial threats" and "how much resources will be additionally required to provide global protection" are of potential interest for individual government when making strategical decisions outweighing cost and effort against global responsibility.

1.4.3 Managerial

The initialization, organization and execution of even regular space missions itself is a time-intensive process with the complex system of systems environment consisting of a vast number of systems. Considering a limited warning time, it is imperative to take the managerial aspect of a planetary defense mission into account. Although technically feasible, due to resource allocation and authority conflicts, a possible mitigation mission might be rendered infeasible when the normal mission preparation time cause the total mission time to exceed the available time, the warning time.

1.4.4 Technological

Since NASA budgets for the expansion plans of threat detection capabilities have been reduce significantly over the course of the past years, it is imperative to identify and quantify required capabilities to meet warning time requirements to launch and execute threat mitigation efforts with current means of technologies, launch capabilities, and mitigation options. Minimum warning times depend on the threat type and corresponding impact probability as well as the available technology to counteract threats. Depending on the extend of the warning time, the need for specific technology development shall be apparent.

1.5 Objective

The objective of this project is the development of a comprehensive system of systems framework to analyze and evaluate mitigation mission alternatives for planetary defense efforts against extraterrestrial threats. By consideration of involved stakeholders and their motivation, current and future detection capabilities, mitigation alternatives and vast variety of threat types, a guide for planetary defense shall be generated. This guide, the so-called "Playbook", contains mission blue print and system requirements, enabling rapid planetary defense in the case of the detection of an actual planetary threat. The Playbook furthermore, shall inform users about required detection capabilities to ensure sufficient warning time to initiate appropriate threat-specific mitigation missions, whereas the mission includes information about required resources and allowable resource allocation and authorization times. A user may utilize the developed Playbook in different ways:

- Determination of minimum detection system capabilities to ensure threat detection that provides sufficient warning times
- Determination of minimum asset and resource availability as well as allocation
- For a given threat and corresponding warning time; provide a mission blueprint utilizing existing technologies and/or guidance on technology development required to counteract the threat

Chapter 2

Background

Asteroids are classified based on a variety of factors, and depending on their classification, mitigation strategies will be very different. One of these ways used to classify asteroids is by their composition, or what materials they are made up of. Asteroids are broken into three different categories based on composition: M-type, S-type, and C-type.

2.1 Notional Planetary Defense Process

The notional process of planetary defense efforts includes several distinct elements as depicted in Figure 2.1. A potential threat is first being detected and the distance from Earth as well as its trajectory is determined. The observation data is then passed to entities that conduct a threat classification in which the size, mass as well as additional threat properties such as material and composition are determined. Based on that data, suitable mitigation alternatives are identified and a corresponding mitigation mission is carried out. Afterwards the effectiveness of the conducted mitigation mission is analyzed and evaluation by tracking the threat's trajectory. In case the mitigation mission was not successful and the threat's trajectory has not been altered sufficiently to guarantee that the object misses/passes Earth, a follow-up mitigation mission must be initiated.

Figure 2.1: Notional Planetary Defense Process

2.2 Property Classification

M-type, or metallic type, asteroids are made up of mainly metals like iron and nickel, and they make up about 8% of all asteroids. An M-type asteroid can be pure metal or mixed with stone, and mostly originated from rocky remnants 4.6 billion years ago, which have now migrated to the asteroid belt. Another property of M-type asteroids is that they have an albedo between 0.1 and 0.2, meaning they absorb a lot of light and appear dark to the human eye [23]. The largest known M-type asteroid is named 16 Psyche, and it has a diameter of 200 km. This size places it in the list of the top ten largest asteroids in the asteroid belt, and unlike any other asteroid discovered, its core is made fully out of metal. Another large M-type asteroid is 22 Kalliope, which has a diameter of 166 km. [24][25]

S-type, or silicate asteroids, are made up of mostly stony materials such as silicon and nickel-iron. The majority of these asteroids are found in the inner asteroid belt, roughly 2.2 AU away from the sun; however, even at 3 AU away from the sun, they are still quite common. S-type asteroids make up 17% of all asteroids, and unlike the other two classifications of asteroids, S-type asteroids are considered relatively bright based on their albedo. In fact, the brightest S-type asteroid is 7 Iris, and it is one of the brightest asteroids ever discovered. Beyond the S-type class, 5 sub-classes go to characterize S-type asteroids even further. A-class asteroids are characterized based on their reddish spectrum shortward and their olivine feature; they are relatively uncommon. K-type asteroids are also relatively uncommon and generally have bluish long wards accompanied by low albedos. L-type asteroids have reddish spectra shortwards as well as a flatter spectrum in infrared as compared to the other sub-classes. Q-type asteroids are characterized by a spectral slope, indicating the presence of metal. Finally, R-type asteroids are moderately bright and have a possibility of plagiocase. [26][25]

C-type, or carbonaceous, asteroids are often dark in appearance due to an albedo of between 0.03 and 0.1, and consist of mainly clay and silicate rocks. They are by far the most common type of asteroid, making up 75% of all asteroids [27]. As for location, most C-type asteroids originate in the outer asteroid belt and are very difficult to detect due to their low albedo. C-type asteroids are also among the oldest celestial bodies in the solar system, and they can often be characterized by a shape like that of a potato. A famous C-type asteroid is 1 Ceres, which is renowned for its size, being large enough to be considered a dwarf planet. [28]

2.3 Orbital Classification

Asteroids are also classified based on their orbits and their various orbital elements. Specifically, NASA classifies asteroids with the following orbital parameters: semi-major axis (a), perihelion distance (p), orbital period (P), aphelion distance (Q), minimum orbit intersection distance (MOID), and absolute magnitude (H). Figure 2.2, shown below, shows how to characterize asteroids using a few of the previously discussed parameters.[29]

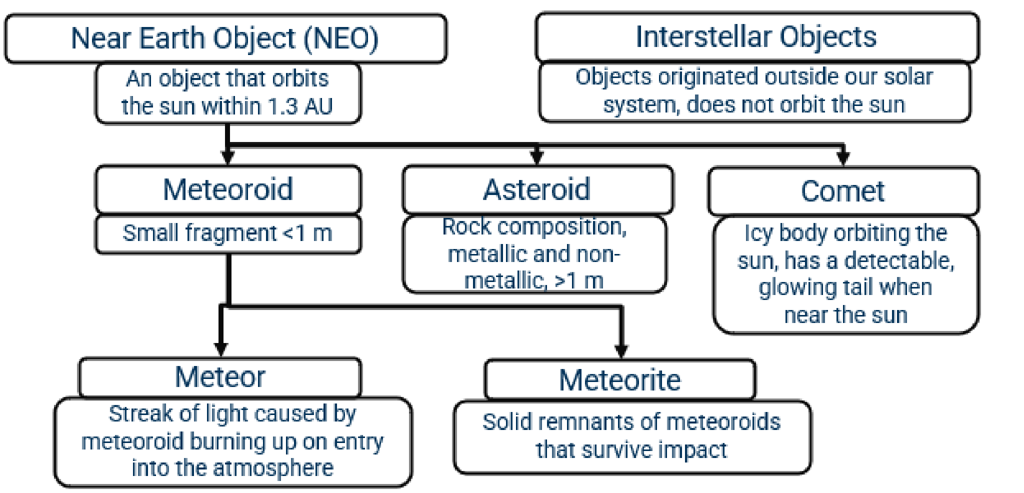

Figure 2.2: A flow chart showing the breakdown of NEO objects [29] [25]

2.4 Threat Classification

In the end, however, what scientists and engineers care about the most is what would happen were an asteroid to crash into Earth. All is well when the asteroids are floating harmlessly in the asteroid belt; however, when one actually makes impact with Earth, what will really happen? This is another way to classify asteroids, based on how much damage they would cause. Damage from asteroids is based on the kinetic energy they have when entering Earth's atmosphere. The kinetic energy equation is shown in Equation (2.1).

$$KE = \frac{1}{2}mv^2 \tag{2.1}$$

This equation is dependent on the mass and velocity of the object, with the more impactful value being the velocity, as this value is squared. In other words, the mass of the asteroid does matter; however, the velocity plays a much larger role in determining the damage the asteroid would cause. It is important to note, most NEO objects enter the Earth at around 25 km/s. This value is relatively consistent across all asteroids, so oftentimes scientists will assume this as an average value and then analyze the impact of different-sized asteroids. Figure 2.3 shows the resulting "type of event" as well as the time between impacts, and the energy for asteroids of different sizes [30]

Diameter of Impacting Asteroid	Type of Event	Approximate Impact Energy (MT)	Average Time Between Impacts (Years)
5 m	Bolide	0.01	3
25 m	Airburst	1	200
50 m	Local Scale Devastation	10	2000
140 m	Regional Scale Devastation	300	30,000
300 m	Continent Scale Devastation	2,000	100,000
600 m	Below Global Catastrophe Threshold	20,000	200,000
1 km	Possible Global Catastrophe	100,000	700,000
5 km	Above Global Catastrophe Threshold	10,000,000	30 million
10 km	Mass Extinction	100,000,000	100 million

Figure 2.3: Probability of different-sized asteroids impacting Earth and the resulting impact [18]

Naturally, asteroids would form massive craters and flatten the landscapes around where they land. However, this is not the only damage an asteroid impact would cause. Asteroid impacts would also cause thermal radiation from the heat they create, which can cause fires and eye damage for those surrounding the impact. Airburst explosions could also release even more energy, possibly equating to the same energy release as from 100s of tons of TNT. Furthermore, atmospheric ionization could also occur. Were the asteroid were to land in the water, which it is likely to do, as most of Earth is covered in water, it could create massive tsunamis, which could also completely level cities. [31]

2.5 Threat Detection

Due to the many classifications of asteroids, there are a few different ways to detect asteroids, depending on some of their characteristics. There have also been projects conducted by NASA that sought to detect and track a large percentage of asteroids. One of these projects to detect asteroids was the Spaceguard Survey, which has detected at least 90% of NEOs greater than 1 km in diameter. In 2009, NASA stated this benchmark goal was close to being met; however, some improvements still had to be made [32]. In 2005, NASA announced the NASA Authorization Act, which sought to detect at least 90% of all NEOs 140 m in diameter or greater; however, by 2019, it is estimated that this value was only around 40% [33].

2.6 Launch Capabilities

The detection tools currently being used to track and detect asteroids are ground-based telescopes, space-based telescopes, and radar tools. Ground-based telescopes focus mainly on visible light, and they are the most common, as Earth already has plenty of ground-based telescopes built and ready to use. The limitation of the ground-based telescopes, however, is that they struggle to find NEOs at a distance or ones that are approaching from the sun. This is because ground-based telescopes can't look at the sun, making this a blind spot for NEO detection [32]. Space-based telescopes use infrared radiation to detect NEOs, and it does so very effectively. The drawback, however, is that space-based telescopes are very expensive to both build and launch, so using this as the only detection method would be impossible [34]. Finally, radar observations rely on radar signals to track NEOs. After the initial detection of the NEO radar signals are sent out in order to discover the physical properties of the asteroid, allowing scientists to classify the asteroid based on composition [35].

There are some detection devices and projects currently being worked on, including the NEO Surveyor, the Nancy Grace Roman Telescope, and the Vera C. Rubin Observatory. The NEO Surveyor is planned to launch in 2027, and the objective is to meet the previously mentioned 2005 goal of detecting 90% of all NEO objects greater than 140m in diameter. This telescope is an example of a space-based telescope, which, as previously mentioned, uses state-of-the-art infrared technology in order to detect satellites effectively. In fact, the NEO surveyor is the first spacecraft built specifically for detecting and tracking NEO objects [36]. The Nancy Grace Roman telescope is planned for launch in October 2026, and it makes use of visible light and albedo values in order to detect NEO objects. Scientists and astronomers project that it is capable of detecting over 100,000 NEO objects as it works with the NEO Surveyor and the Rubin Observatory to improve the measurements of NEO objects by two to three orders of magnitude [37]. Finally, the Vera C. Rubin Observatory has been operational since May 2025, and it is an example of a ground-based detection system. As with most ground-based systems, it focuses on the visible light emitted to detect and track NEO objects. It is forecast to be capable of detecting 127,000 NEO objects and be great at the initial discovery of NEO objects [37][38].

2.7 Past Threats

An Armageddon scenario, however, is not as far-fetched as some may think from occurring. In fact, there have been an abundance of cases where asteroids have already hit Earth, or where scientists have believed an asteroid may contact the Earth's surface. In 1998, astronomers thought that a NEO named XF11 would hit Earth in 2028 [39], and then, more recently, in December of 2024, astronomers believed a NEO named YR4 was going to make contact with Earth in 2032 [40]. Both XF11 and YR4 were later determined not to be a threat; however, these scenarios highlight the importance of being more vigilant about NEO objects floating around the solar system.

Furthermore, the urgency of rapid planetary defense capabilities, mitigation alternatives as well as required detection capabilities, is underlined when considering Table 2.1 showing the most recent detections (as of October 16th) of asteroids that closely missed Earth. The presented excerpt of close fly-bys of asteroids which have a diameter of at least 20m and a miss distance of approximately 50,000 (statue) miles or less, shows that with current detection capabilities and detection volume, warning lead times of less than a day are rare. Those Asteroids with diameters around 20 m—such as the Chelyabinsk impactor in 2013, which produced an explosive energy release of roughly 500 kT TNT equivalent—already pose a significant threat. Objects like 2025TQ2 and 2025TF, however, are estimated to be at least 30 m in diameter and could reach megaton-level yields. For objects in this size class, no practical mitigation option exists within the available warning time, not even the deployment of nuclear warheads. An impact of such an asteroid would likely destroy an entire city or even a metropolitan region, with no realistic opportunity for human intervention.

Table 2.1: Hazardous Objects – Diameter >= 20 m, Miss-Distance <= 50,000 sm

Identifier	Detection Date	Date Closest Approach	Distance [sm]	Diameter [m]
2025UJ4	20-Oct	21-Oct	38,000	30
2025TC	1-Oct	3-Oct	53,000	27
2025TQ2	2-Oct	2-Oct	7,000	31
2025TF	1-Oct	1-Oct	4,200	32
2025SU4	23-Sep	24-Sep	14,000	32
2025SU7	24-Sep	24-Sep	45,000	31
2025SY1	18-Sep	19-Sep	36,000	32

2.8 Threat Mitigation

The scientific community has proposed a variety of different mitigation strategies, mainly based on two different fundamental principles, so called "push/pull" and "ablation/vaporization" strategies. The effect of both principles however, is equivalent. Both principles induce a momentum change of the threat which causes a deflection from its original trajectory. Threat mitigation therefore does not aim for the actual destruction of extraterrestrial threats rather than deflecting them. However, based on mitigation approach and the size as well as the composition of the threat, destruction of the object may be a side-effect of the mitigation, deflection efforts. In fact, unintentional destruction of an object may have severe repercussions, e.g. if fragments remain on an Earth-bound trajectory while being of size and composition that the fragments will enter Earth's atmosphere in form of an asteroid shower, potentially causing multiple impacts or explosive mid-flight disintegrations and thus widespread and severe devastation. Fractionating may occur intentionally, but requires subsequent mitigation missions which have a significantly shortened warning time. Although several different approaches were published and analyzed only one concrete mitigation mission has been conducted and proven to be effective so far. The mitigation strategies listed below, can be employed independently or collaboratively, however, the applicability of mitigation strategies highly depends on the threat type, i.e. the size and composition, as well as the warning time, i.e. the time till impact. However, generally speaking, the earlier a threat is detected the easier and safer any mitigation strategy can be executed.

2.8.1 Push/Pull Strategies

For so called push/pull strategies, one or several spacecrafts are required to travel to and directly interact with the threat object to establish the desired deflection of the object from its original Earth-bound trajectory. Depending on how early a threat is being detected the time of flight to the object may be significant, on the order of magnitude of several years, during that time the Earth-threatening object propagates on its impact trajectory [41]. That subsequently implies that a mission failure drastically increases the planetary threat level while reducing the number of feasible follow-up contingency mitigation mission concepts.

Solar Sail Deflection For the solar sail deflection alternative, a spacecraft would attach a solar sail to the object. The sail is then exposed to solar winds emitted by the sun, establishing a propulsive force through the solar wind's radiation pressure acting upon the area of the sail [42]. Depending on the object's trajectory the propulsive force induces a momentum change of the object that causes a deflection from its original trajectory. Since the intensity of solar winds and thus the radiation pressure decreases with the 2nd power of the distance to the sun, the effectivity of the solar sail approach highly depends on the location and trajectory, with respect to the Sun, of the object. Furthermore, the integrity and subsequently the effectivity of the solar sail may be compromised by debris impacts. Since radiation pressure is generating a comparably small force per unit area, large area sails are required and the mitigation effort will take, depending on the mass and trajectory of the object, several years.

Deployable Propulsion Modules For this strategy, spacecraft engines would be mounted onto the object's surface. Their operation induces a force that changes the objects velocity vector, effectively changing its trajectory, therefore causing a deflection [42]. Although current propulsion systems could already be utilized, this strategy is severely limited by the available propellant quantity. However, by providing and/or resupplying additional propellant during operation or through the continuous development of chemical, electrical and nuclear propulsion technology, this approach could become increasingly feasible.

Gravitational Tractor/Tug The deployment of a gravitational tractor or space tug presents a rather futuristic option to cause a trajectory deflection. A spacecraft would be required to hover close to the object, so that the slight but existing gravitational interaction between spacecraft and object cause a trajectory deflection [42]. Since the gravitational force proportionally scales with the mass of the spacecraft and inversely scales with the mass of the object, an extensive hovering time may be necessary and therefore significant fuel resources would be required to maintain the required hovering distance since the object's gravitational force applies a pull onto the spacecraft as well.

Kinetic Impactor The kinetic impact mitigation alternative is the only one that has actually been tested so far. NASA's DART mission has proven that the deflection of an asteroid can be induced by an high-velocity, and thus high-impulse, impact of a spacecraft. The impulse transfer through the impact causes a change in the object's momentum and thus a deflection [14]. Compared to the mitigation strategies presented before, the resulting trajectory deflection caused by a kinetic impactor is almost momentarily detectable. Within the scope of the DART mission the impactor weighing 580 kg and having a relative impact velocity of 6.1 km/s caused a significant change in the asteroid Dimorphos which has diameter of 780 m and a mass of approximately 5 billion kilograms [43][44]. In terms of future mitigation mission utilizing kinetic impactors, this success implies that existing technology, i.e. satellites, could potentially be utilized in planetary defense efforts. Since satellite production of satellites in a comparable mass class, entered a mass-production stage in order to provide assets for satellite constellations, and launch capabilities scaled up accordingly as well, it seems plausible that impactor constellations may be utilized for planetary defense efforts [45]. Impactor constellations could provide fail safe capabilities in case the object fragments, or in case an individual impact does not result in sufficient trajectory deflection [41]. A cause for insufficient deflection may be unreliable information of the objects composition, size, mass and internal structure. The effectiveness of kinetic impactors is significantly reduced when the objects has a porous structure.

2.8.2 Ablation/Vaporization Strategies

Deflection strategies utilizing ablation and vaporization principles induce a reactive force according to Newton's third law to achieve the desired change of the body's trajectory. By application of heat to the objects surface material melts and vaporizes, thus mass is expelled which causes a reactive force in the opposite direction [42]. Generally speaking, ablation and vaporization driven mitigation approaches require application of heat over a long period of time, i.e. several years to show the desired effect on the trajectory.

Laser/Directed Energy The laser or directed energy approach presents an earth- and/or space-bound approach to locally induce heat onto the object's surface by utilization of high-power lasers [42]. A major advantage of this approach is that if adequate lasers are available, mitigation efforts could be initiated instantaneously. Potentially limiting the effectiveness of this approach is however the required pointing accuracy of a laser to consistently expose the desired location on an object's surface. This furthermore requires accurate knowledge of the object's trajectory and its spin and tumbling.

Albedo Manipulation For an albedo manipulation driven approach a spacecraft would need to locally apply coating of the object's surface which locally changes the optical characteristic of the surface to effectively absorb a greater amount of heat from the solar radiation [42]. A standalone albedo change driven approach however presents a comparably unattractive alternative, since the required coating could potentially accumulate to an enormous payload weight, the spacecraft would be required to rendezvous with the object to apply the coating, before any vaporization could induce a trajectory deflection whereas vaporization driven strategies require an extensive period of time to show effectiveness. Finally, the amount of heat that could potentially be absorbed is inversely proportional to the 2nd power of the distance to the sun, rendering this approach location and trajectory dependent.

Nuclear Detonation A nuclear detonation is considered to be the most suitable mitigation option against larger objects when a limited warning time is given. The extreme heat generated by the nuclear detonation would vaporize large quantities of the object, thus induce a significant force deflecting the object [41][42]. Furthermore it is, depending on the object's size, composition and internal structure, likely to cause fragmentation. As mentioned beforehand however, secondary mitigation efforts may be required if the fragments were to large to vaporize and disintegrate during entry into Earth's atmosphere without posing a threat to the population. The distinct advantage of a nuclear option is the availability of nuclear warhead that could be potentially utilized for planetary threat mitigation efforts. However, this approach presents inherent risks and authorization limitations, since weapons of mass destruction would need to be launched into space on launch vehicles that were never meant to launch and transport such payloads.

2.9 Stakeholder

The stakeholder question is difficult to answer in context because there are so many moving variables. However, it can be narrowed down to three main parties that are involved. The first is industry; this is the player who will be the main point of focus for the manufacturing of components. They may not make the decision or know the science behind what is being manufactured; however, they will be the main point of contact for having things built. The shot caller will be the government; they are the ones who will determine whether or not the threat warrants any sort of action. This also includes if there is a threat to a different country, it will be the government that works together and makes the final call on what needs to be done. Finally, the brains behind it, and the ones that will provide the numbers and statistics to the government, are academia. Academia will also provide the industry with data analysis, technology research, detection capabilities, and classification, such that appropriate countermeasures can be manufactured. In turn, the industry will then receive from the government information on the laws and politics, as well as the funding for their mission. Questions to be asked and answered in this relationship are ones such as, do we support the defense of the entire planet or only our allies? Who pays for the mission? Can companies profit from a mission? Similar questions must also be answered between academia and the industry. What industry limitations exist when planning and executing a mission? Can data acquisition be privatized? Who makes the final decision on what mission to run? As is clear, there are many questions and issues that arise when working with many entities to protect the planet. Answering and continuously updating these questions before a threat is detected is vital to a rapid response.

Chapter 3

Architecture

The task of mitigating the threat to the Earth is a complex system of systems problem. In order to create the playbook, a breakdown of the requirements for the playbook and the overall architecture must be understood. The requirements of the playbook include the necessary inputs to the playbook and the necessary outputs to execute a mission. The architecture of the system consists of the individual systems, including hardware, software, and personnel working together to accomplish the mission.

3.1 Requirements

The idea behind the playbook is to have a list of predetermined responses to a threat against the Earth. In order to create a response to a threat, several parameters of the threat must be known before the mission can be planned. In addition, technology and available mitigation options must be evaluated along with the threat parameters. These two categories, threat parameters and available mitigation options, make up the required inputs for the playbook. Since the goal of the playbook is to generate mitigation plans based on input parameters, the output of the playbook is a plan regarding the mitigation method, launch sites, and other pertinent information from the mission. A requirement diagram can be seen in Figure 3.1.

3.1.1 Inputs

There are several parameters required to plan and execute a mitigation mission. First, the system needs to be able to detect a potential threat as far away as possible. Next,

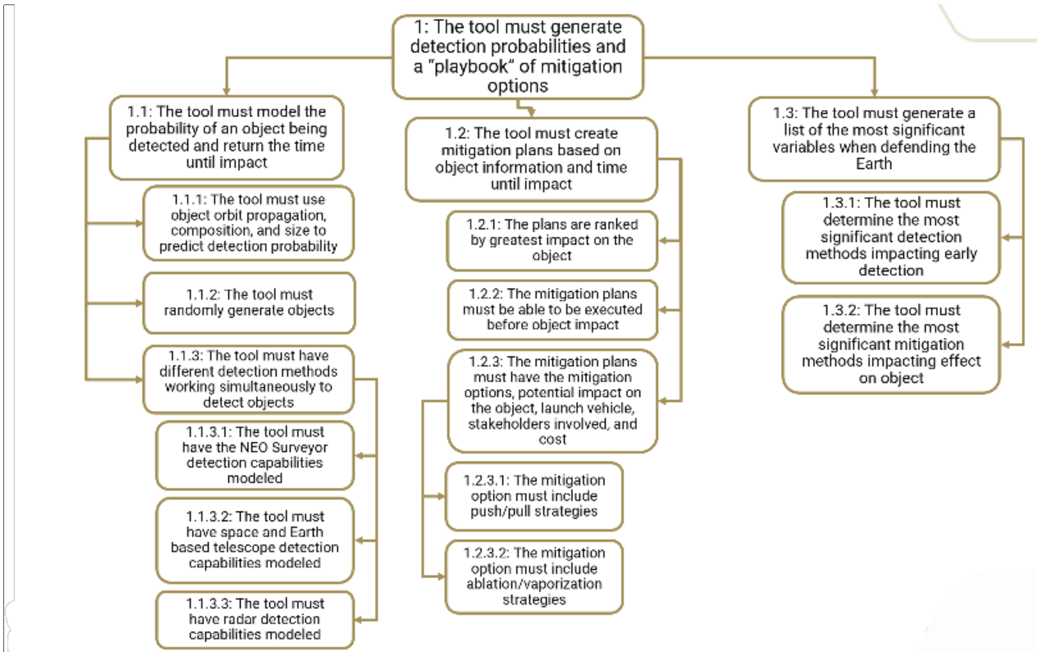

Figure 3.1: Visual representation of systems involved in architecture

the system needs to track the threat and identify key properties. The system also needs to classify the threat to help decide the best response. Then, the system needs to communicate to key stakeholders to select a mission and prepare the mission. Finally, the system needs to execute the mission, including continuous tracking of the target, assisting the mission in guidance and navigation, and surveying the effect of the mission to determine whether it was a success of if another mission needs to be launched.

Based on the needs for the mission, the first input needed for the playbook is time to impact. The time to impact will help determine how long the mission planning phase can be. This is important as the time constraint will determine what technologies are available for this mission. If the threat is predicted to hit in the next few months, then only current technologies can be used and it will be difficult to manufacture an entirely new vehicle. However, if there are years until the threat is predicted to impact, new technologies specifically designed for the specific threat is possible.

The second input needed is the classification of the threat. This includes the speed, size, composition, and orbit properties. This is important as the speed and size determine the kinetic energy, which is indicative of the damage it can cause to the Earth. Additionally, the possible mitigation techniques may vary depending on the size. If a threat is too large and too close, a kinetic impactor may be useless, so another option has to be deployed.

Furthermore, the size and distribution of mass would affect the location on the threat where a mitigation mission would aim for. The composition helps determine what mitigation techniques are most effective, as different properties react differently, and different porosities would change the technique needed. Finally, the orbit properties such as rotation can assist in the planning and navigation of the mission.

Once the target is analyzed, the available mitigation techniques must be accounted for. The mitigation techniques vary on a few factors. First, the time frame of impact is important as it affects the technologies available for the mission. As mentioned previously, the closer the threat is, the less time new technologies can be research or manufactured. Second, the mitigation techniques are dependent on the stakeholders involved. If government entites decide that because the threat may not affect their specific country, they may decide to not contribute to the mission. This would result in less funding and different options. Additionally, different private companies would have to invest in research and development for a mission that has no actual profit, so their participation or demand of captial would affect the mission. Finally, if the time frame for the threat is short, then available products from current companies must be scrambled and put together. All these decisions involving the mission and stakeholders also leads to the question of who makes the decisions in this situation, so the area predicted to be impacted the most would be a requirement so stakeholders who are closest to the impact zone would have a greater say.

3.1.2 Outputs

As mentioned previously, the goal of the playbook is to generate plans for different threat scenarios, so the output of the playbook must be the mission design. For a successful mission, the mitigation technique and mission plan must be created. The mitigation technique will ultimately be decided based on the inputs mentioned previously. The techniques were also discussed earlier, but more could arise in the future. The mission plan is a detailed plan of the launch timing, vehicle, impact location, tracking and navigation plan, expected effect, probability of success, and the associated stakeholders and their role in the mission.

The launch timing is a result of the orbit and projected location of the threat. It also depends on the stakeholders involved who have launch sites and the time they can get the launch ready in. The launch vehicle is the rocket that will take the mitigation technique out of the atmosphere. The vehicle will be determined based on the size of the mitigation technique. The impact location is the projected location on the threat the mitigation technique will be deployed. This is a factor of the size distribution and rotation of the threat. For example, if the mission is to change the rotation of the threat, the location may be the edges of the threat. The tracking and navigation for the mitigation technique to reach the threat is also important. It may require different satellites or telescopes to be re-tasked to assist the navigation. The expected effect is what the mission is trying to accomplish. It may target a change in the targets orbit or splitting the threat into two parts. The method of analyzing the effect of the mission must also be outputted. Finally, the probability of success must be calculated. There is never a 100 percent chance of success, and the probability of success may determine whether a mission plan would be selected. A mission could be theoretically extremely effective, but if the probability of success is low, that plan may be ignored or it may be selected contingent on backup plans being ready.

3.2 System of Systems

To satisfy all the input requirements and accomplish all the outputs, many interconnected systems will need to work together. This complex network of systems includes satellites, telescopes, different vehicles, stakeholders, and data analyses. Figure 3.2 gives a visual representation of the different systems working together. The architecture has two parts: the roles of the individual systems and the interactions and flow of information between the systems. Each system and interaction between the system must contribute to satisfying the requirements mentioned earlier.

3.2.1 Individual Systems

First, the systems that will satisfy the input requirements will be defined. The first input to the playbook is the time to impact. In order to find the time to impact, the threat needs to be detected. As mentioned previously, threats are detected using satellites and ground-based telescopes. The main system responsible for detection will be the NEO Surveyor, as its mission objective is to detect NEOs. It is the only satellite that is tasked mainly to detect threats. Additionally, its use of infrared radiation detection is the most effective way to detect objects in space. Other satellites that detect objects and potential threats in space have alternative missions, so they are grouped together under general satellites. These satellites vary between infrared and visible light telescopes. One satellite telescope

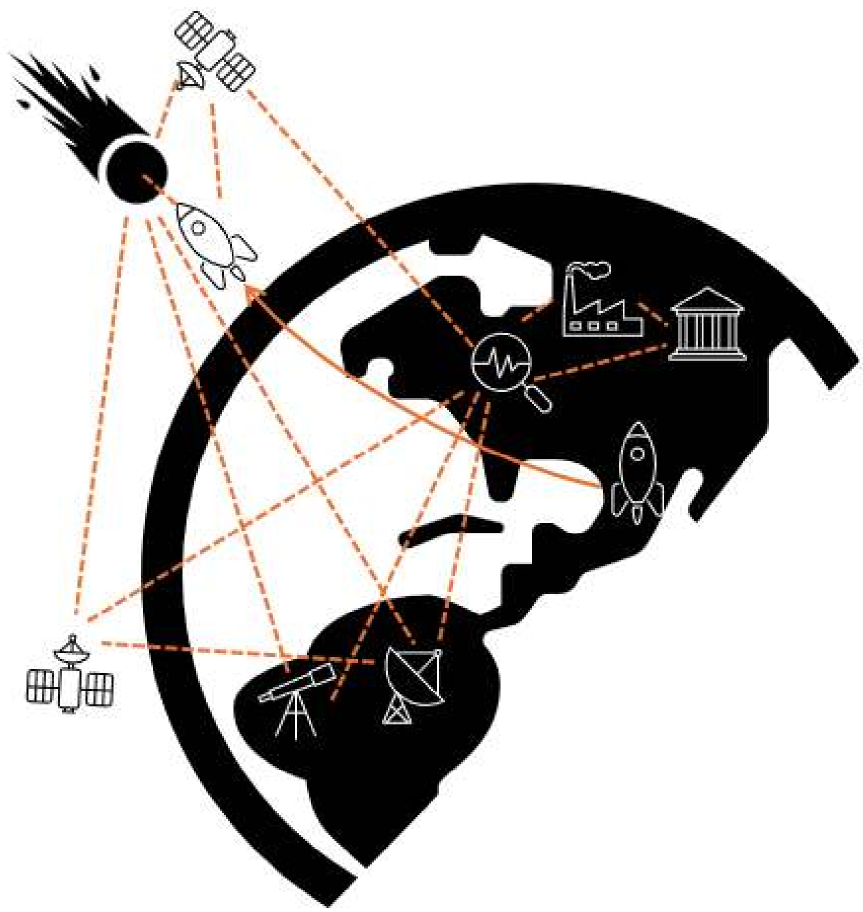

Figure 3.2: Visual representation of systems involved in architecture

that headlines the general category is the Nancy Grace Roman Space Telescope. Although its main goal is not to discover NEOs and threats, it will be effective in assisting the NEO surveyor. Additionally, there are ground-based telescopes. The Vera C. Rubin observatory is an example of this type of telescope. These systems are effective at making the initial discovery and threat detection of objects in space. Next, the threat needs to be classified by its speed, size, composition, and orbit properties. The system responsible for this are the ground-based radars. These radars effectively classify the threat if the threat can already has already been detected. Some examples of these radar systems are the Goldstone and Arecibo radar dishes. The range of the radars are limited to interplanetary ranges, so if the playbook deems it necessary, additional funding and research may be required to extend the range of systems that can classify the threats. Once the target is detected and analyzed, a system is required to evaluate potential mitigation option. To evaluate the mitigation options, analyses is needed, either from a research institute, government organization, or private corporation. This requires funding and technology. Depending on the time frame and the technology selected, it would also require more research. After the mitigation technique is selected, the other outputs are accounted for by various stakeholders. The stakeholders include research institutes, private corporations, and governments. The governments fund the mission planning and mission itself, while research institutes and private corporations analyze the mitigation options, plan the mission, launch the mission, and evaluate the effects. When the mission is actually, an additional system of the launch vehicle is necessary. This vehicle gets the mitigation system out of Earth's atmosphere. Finally, the last system is the mitigation technique. This system is responsible for delivering the actual mitigation technique to the threat.

3.2.2 Concept of Operations

In any system of systems, the concept of operations (CONOPS) needs to be clearly outlined so all the systems work efficiently together. The first step in an affective CONOPS is outlining the flow of information. Figure 3.3 shows what type of information is transferred between the different systems. As seen in the figure, data, resources, and funds are transferred between different systems. Next, the order in which operations are carried out must be defined clearly so missions can be deployed quickly without delays. The flow of information begins with the satellites and telescopes responsible for the initial detection of

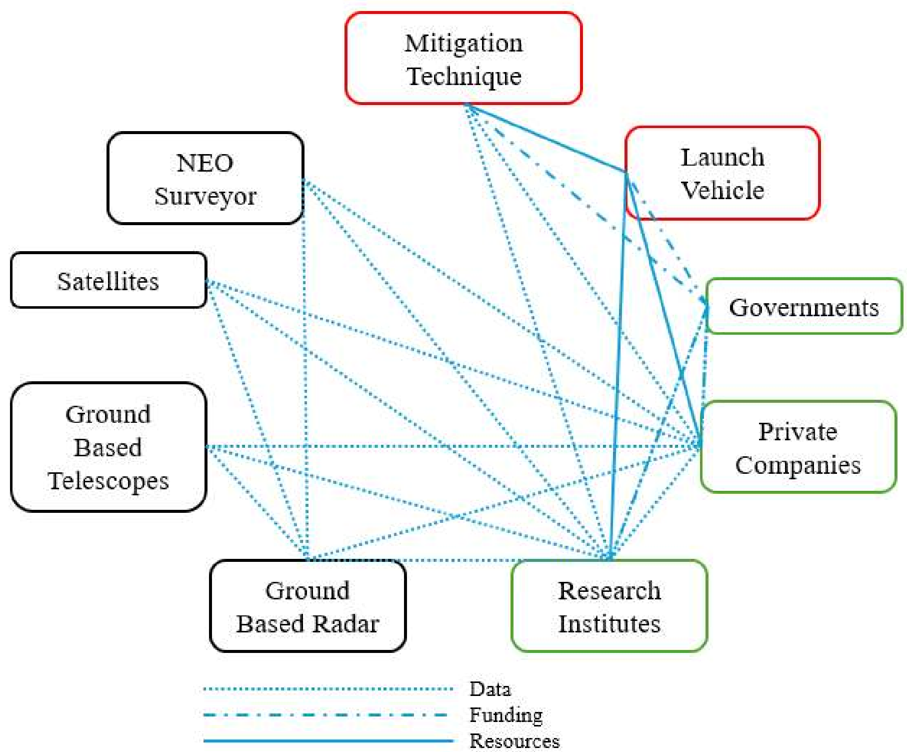

Figure 3.3: Flow of information between systems

the threats. Once the data on the threat is collected, it is send to research institutes and private companies for analysis. These two stakeholders would begin the process of selecting and planning the mission. Meanwhile, the data and potential missions are relayed to the government who would fund the mission and make the final decision on what mission should be used. Next, the research institutes and private companies pour resources into the launch vehicle and mitigation technique. Finally, the vehicle is launched and with the assistance of data from the stakeholders and satellites, the vehicle can reach its target and execute the mitigation mission.

The CONOPS for this system of systems has two parts to it. First, the playbook of possible mitigation techniques are made. These missions are made in advance based on a variety of factors. The playbook would contain sample missions for different scenarios based on the inputs mentioned previously. It would evaluate what options are possible for what type of threat and the time frame of the threat. Additionally, the sample missions would be modeled and simulated, allowing for better preparation and understanding of responses to the threats. These simulations would inform us of the necessary detection capabilities necessary. For example, if the simulations reveal that for a threat bigger than 50 meters, the threat must be detected at least a month in advance, then the detection systems would have to be improved to meet that capabilities. So while the system of systems has some requirement inputs to make the playbook, the playbook would give an idea of what the technical requirements of the systems are, including detection time, launch time, funding and more. While the playbook is being developed and improved, the systems are already operational, scanning space for potential threats. When a threat is detected, the two parts coincide as the sample missions now become real missions that must be executed.

Chapter 4

Modeling

The goal of the model created is to take the inputs mentioned previously and generate the desired outputs so a playbook can be compiled. The model can be split between the detection modeling and the mitigation modeling. The overall workflow of the model can be seen in 4.1.

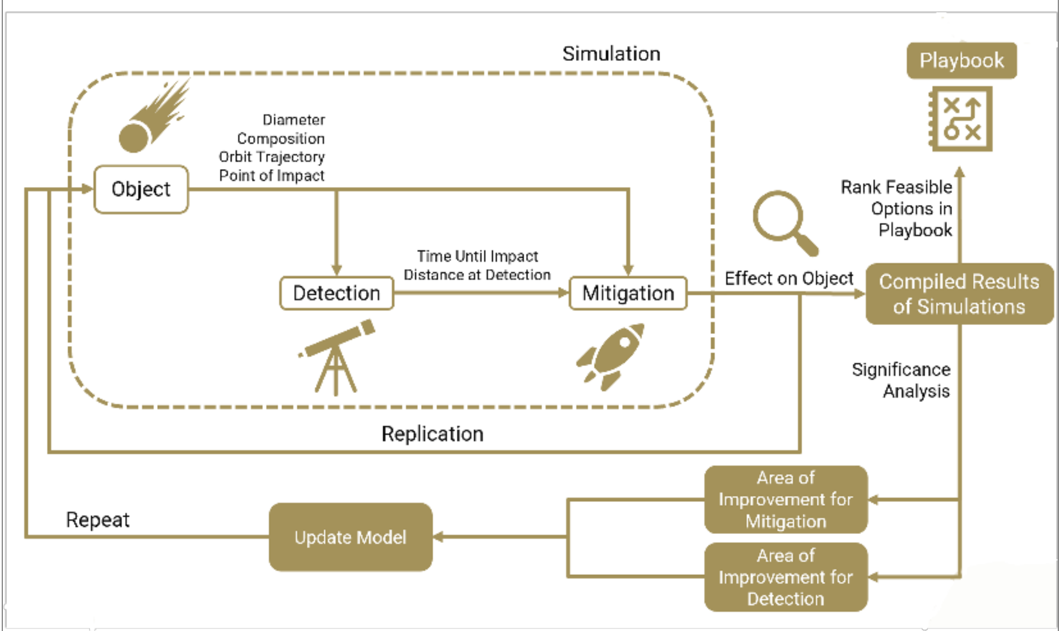

Figure 4.1: Overall Flow of Modeling

A threat is generated with random physical and orbital properties and fed into the detection modeling. The detection modeling calculates a distance at which the threat is detected, which is then fed into the mitigation modeling. The mitigation modeling generates effects due to different mitigation techniques, and compiles them into the playbook. This process is repeated thousands of times to accumulate a list of possible threats across different physical and orbital properties to identify the weakest points of defense. The model can then generate the variables of detection and mitigation that can improve the defense of the planet the most. The improvements can then be added to the model, and the simulations performed again so the playbook can be recreated and the defense of the Earth can be evaluated. The potential options the playbook is based on can be compiled in a morphological matrix as seen in 4.2.

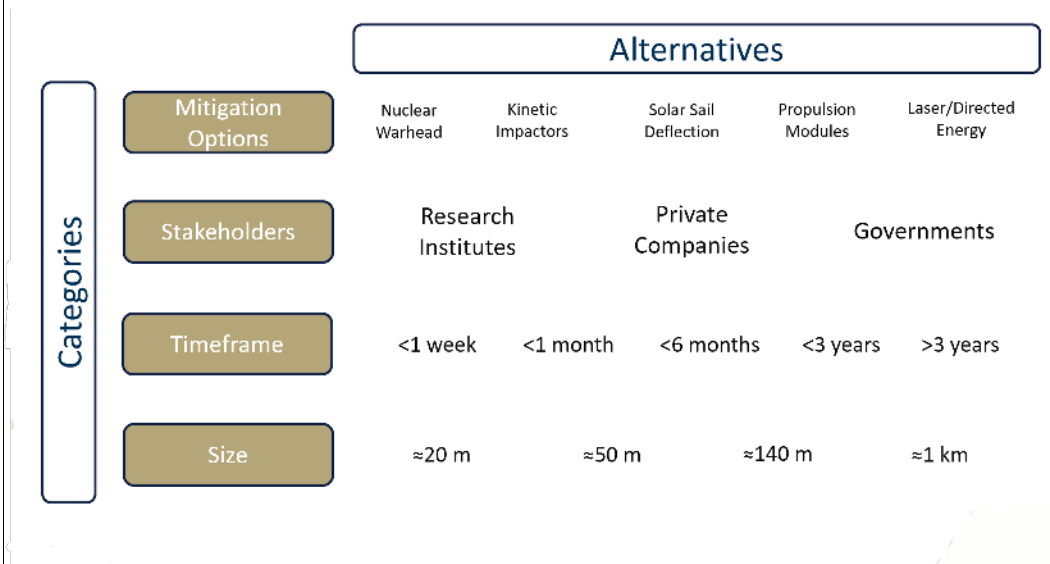

Figure 4.2: Matrix of Alternatives for Mitigation Options

4.1 Detection Modeling

The detection modeling is created using a custom-made code in Python. There are three parts to the detection modeling: threat modeling, detection systems modeling, and point of detection modeling. The three parts work together to output a point at which the threat can be identified with a percent confidence specified by the user. During the detection modeling, it is assumed that the orbit trajectory is affected only by the Sun, making it a 2-body problem, and that the object is perfectly spherical.

4.1.1 Threat Modeling

The threat modeling is the first part of the overall modeling sequence. This portion of the modeling determines the physical and orbital properties of the object being detected. There are several threat parameters required for the threat. First, the threat's size and composition need to be determined. The size can range from 10 meters to 1 kilometer, while the composition is randomized by albedo. Once the albedo is determined, the object is classified by its type if its asteroid or if its a comet. This is important in determining its visibility by the detection systems and the effect the mitigation techniques will have on the object. Next, the orbit of the threat is determined. The modeling first begins by identifying the point and angle of impact on the Earth in the Earth-centered frame. This is converted to the sun-centered frame and position of the threat is propagated backwards in time using its velocity vector until the threat is 5 years away. The position of the threat is then calculated at different time intervals and is then fed into the detection systems modeling, which is then used to find the position and thus the time until impact when the object is detected. The position is used to determine whether or not the threat is even in the scope of a detection system.

4.1.2 Detection Systems Modeling

There are 3 different detection systems that need to be modeled: infrared telescopes, visible light telescopes, and radar systems. There are several parameters for each system required for the modeling. First, there are the basic equations for detection. These equations determine the signal of the threat detected by the system. Infrared telescopes detect objects using the radiation emitted from the object. It requires the temperature of the object and the spectral radiance. The temperature of the object is estimated using the Near-Earth Asteroid Thermal Model (NEATM) while the spectral radiance is estimated using Planck's Law. Additionally, the wavelength bands that the infrared telescope detects is also a required input. Finally, the background noise detected by the infrared telescope must be quantified so a signal-to-noise ratio can be calculated, which is then turned into a probability of detection. The basic detection equation for a visible light telescope is the apparent brightness equation. Similar to the infrared telescope, the background noise (other visible light) must be quantified so a signal-to-noise ratio can be calculated. Radar systems calculate detection using the monostatic radar equation.

Apart from the basic detection equations, the position and field of view (FOV) of the systems must also be known. For the threat to be in the scope of the system, the position of the threat relative to the system must be known. In order for the system to even have a chance at detecting the threat, the threat must be located in the FOV. This is calculated using the incidence angle with the Sun, which is calculated as the angel between the Sun and threat from the system's point of view. The position of the system relative to Earth's center must also be known.

4.1.3 Point of Detection Modeling

The point at which the threat is detected is calculated as the point at which there is 95 percent chance of detection. For every timestep, the probability of each detection system detecting the threat is calculated and combined using the probability of a failure. This means each probability is subtracted from 1 and multiplied together. The resulting value is the percent chance of the systems NOT detecting the object. This process is computed every timestep and multiplied together until the percent chance of not detecting the object is 5 percent, meaning the percent chance of detection is 95 percent. This time until impact and distance to Earth is calculated at this point and then used as an input to the mitigation modeling.

4.2 Mitigation Modeling

Mitigation modeling consists of two custom-made Python script sections. The first one focusing on the actual mitigation effect achievable by given mitigation approach alternatives, timelines and threats, threat types. Since the threat type and furthermore composition determination is based on optical observation and analysis, and is therefore inherently with uncertainty afflicted, the goal is to identify technically feasible mission scenarios that result in a mitigation success probability greater than a user-defined minimum certainty threshold. In a second step the theoretically technically feasible mission scenarios will be further analyzed to determine if actual technical feasibility is achievable. For all mitigation approaches that require a launch and transportation system the required energy for launch, flight and if applicable maneuvering will be determined. Based on the calculated metrics technical feasibility will be investigated. For ablation/vaporization based mitigation approaches mitigation approach (additional) specific metrics will be utilized for the feasibility evaluation.

Finally, an option will enable the consideration of the technical complexity/risk of an intended/chosen mitigation mission approach by incorporation of a complexity factor, which depends on subject matter knowledge and intuition. For an example, a nuclear warhead mitigation approach poses less risk for mission failure due to mission complexity than an approach utilizing a solar sail or propulsion modules which must be mounted onto the threat and therefore require a rendezvous maneuver as well as direct and most likely autonomous operation in close vicinity of the object over an extended period of time. This "qualitative" mission complexity/risk factor enables a probabilistic evaluation of mission success. A mission is considered to be successful if the mission success probability is above a user-defined threshold and the closest approach, the so-called miss distance is equal or greater than a user-defined threshold.

Mitigation Effect Modeling

In this initial step of the mitigation modeling, available mitigation approach alternatives are applied in regular and continuous time increments along the threat's trajectory. Starting at the time of detection when the time till impact, the mitigation time frame, is known it is assumed that the mitigation efforts can be applied directly; at this stage the required time of flight or time till mitigation effect application is not considered. For set parameter ranges for the mitigation alternatives: single/multi kinetic impactor (push/pull), solar sail deflection (push/pull), deployable high- and low-thrust propulsion modules (push/pull), laser/directed energy (vaporization/ablation), and nuclear detonation (vaporization/ablation) the mitigation effectiveness is calculated. The mitigation effectiveness is based on the first order astrodynamics mechanics. Since all mitigation approaches, besides the nuclear detonation, change the velocity vector of a threat by changing the total threat's momentum either instantaneously, e.g. for the kinetic impactor or according to the assumption for high-thrust deployed propulsion modules, or continuously for the low-thrust deployed propulsion modules, solar sail and laser/directed energy approach, the updated post-mitigation trajectory of the threat can be computed. For mitigation approaches that apply a force continuously onto the threat, the threat's trajectory must be integrated over the time period over which the force is applied to account for the "infinitesimal" affect on the trajectory for a given timestep. Finally, for a given threat trajectory, list of lists are created. For each investigated point in time along the initial threat's trajectory, for each mitigation approach, a the possible design space for the specific mitigation mission parameters is calculated and saved for further processing. By choosing the design parameter ranges for each mitigation mission realistically this yields the first results whether or not successful deflection with desired success probability is achievable. Only successful mitigation mission setups are then passed into the 2nd phase of the mission feasibility analysis.

Kinetic Impactor Mitigation mission parameters for the kinetic impactor alternative are the impactor's mass and the relative impact velocity and the impact angle with respect to the threat's initial velocity vector orientation. For mitigation efforts utilizing a constellation of kinetic impactors, two main concepts of operation are assumed, the simultaneous and sequential impact. The simultaneous impact of multiple kinetic impacts is hereby representative for the impact of a massive single impactor for a given relative impact speed and impact angle. For this case it is assumed that the relative impact velocity and mass of all impacts is the same. In case of the sequential impact mode, kinetic impactors of the same mass impact the target, the asteroid, in with a user specified temporal difference.

Deployable Propulsion Modules For the deployable propulsion module approach, once again two operation concepts are investigated. The high thrust burn, assumed to be an instantaneous maneuver, and the low thrust burn. For both, the available propulsion force is the design variable. Furthermore, the thrust vector direction is another design parameter for the deployable propulsion module mitigation approach.

Solar Sail The solar sail mitigation approach features 3 distinct design variables which are varied in the a-priori user-defined value range. Those values are the area of the solar sail, the absorptivity coefficient of the sail's material, and the time-specific damage coefficient. Furthermore, besides the threat's trajectory, the distance between threat and Sun must be computed along the threat's trajectory. Additionally, the location of the threat with respect to the sun is determined to allow for correct pointing of the solar sail to effectively capture and take advantage of the solar winds, that then create a resulting force which is applied to the threat.

Laser/Directed Energy A laser/directed energy ground-based mitigation approach applies a consistent force on to the incoming threat. The created reactive force is proportional to the laser power that reaches the threat and the beam diameter of the laser. Furthermore, the threat's surface/composition characteristic is accounted for that has a major impact on the ejection velocity of vaporized material which then creates the reactive force onto the threat object.

Nuclear Detonation For the nuclear detonation option, which generally is part of the vaporization/ablation category mitigation approaches, only a check if mitigation is generally possible can be conducted due to the lack of publicly available data on the vaporization effect of nuclear detonations. Therefore, based on during the literature research identified energy yield classes that indicate the required warhead explosive yield in order to destroy an asteroid threat, it is only checked whether or not a warhead has enough potential yield to destroy an object. In this special case, the only design variable for the mitigation approach is the yield of the warhead and the number of warheads utilized. It is assumed that if multiple warheads are utilized that they are deployed distributed across the threat object's surface.

Technical Feasibility Modeling

The initial calculation data in form of a list of lists from the previous step is now investigated regarding the actual technical feasibility. For the mitigation strategies that require a launch and transport of payload to the threat the required characteristic energy to escape Earth's gravitational field and the to launch and maneuver is compute to first order accuracy. Since all mitigation strategies were applied at given time increments along the threat's trajectory, the time of flight and the orbital state vector of the threat is known, therefore simplified trajectory determination is conducted. The total payload mass of the mitigation equipment, required to accomplish the successful threat trajectory deflection is computed based on the conceptual mission data from the first step. Accounting for the fuel mass required for the flight and (if required) maneuvering, and the determined payload, the required launch capability can be approximated and compared with currently available launch capabilities. As proxi for the launch location an average latitude between Vandenberg Space Force Base in California and Cape Canaveral Space Force Station in Florida. For the ground-based laser-based mitigation approach technological feasibility is evaluated based on the total ground power required to provide the laser power at the threat's surface accounting for transmission losses and currently available laser systems as well as more generically the required infrastructure to provide the ground-based laser power.

Commercial software applications and available open-source Python libraries have been investigated and evaluated regarding their possible application for both the mitigation effect modeling as well as the launch capability hence trajectory-based time of flight computation. The most prominent commercial product, the Ansys System Tool Kit (STK), although accessible and automatable by custom Python scripts, requires extensive computation time to compute trajectories and mission profiles. Since running of several thousand cases is required for this project, STK is not a feasible modeling and simulation option. Therefore, specialized Python libraries are preferred that provide equivalent astrodynamical computation capabilities. "Pykep" and "Poliastro" are identified as potential Python libraries with required build-in capabilities. By creating a fully custom-build Python script for the two-step mitigation approach modeling it would be possible to streamline the computation to promote time-efficient analysis enabled by multiprocessing to even further decrease the required computation time. Furthermore, using Python libraries instead of a commercially available tool promotes direct controllability regarding the fidelity level of the analysis and increased flexibility for potential code adjustments and expansions.

4.3 Modeling Improvements

Once the detection and mitigation modeling for thousands of cases has been completed and the results are compiled in a playbook, the playbook can be evaluated for weak spots and suggestions for improvements to the system of systems can be made. First, the inputs to each mitigation method model are analyzed to find the most impactful variables. The inputs are varied to simulate improvements and then determine what improvements result in the biggest impact. Additionally, the detection modeling systems can be varied and tested in the same way to determine what detection system improvements yield the best results. This is an important part of the modeling sequence as the overall tool can be used to make suggestions as to what technologies to invest in to keep the planet safe. If detection time is the most important variable, then the detection system should be invested in, which means the exact system and parameters of the system can be estimated to yield the best results.

Bibliography

[1] A. Mainzer, "The future of planetary defense," *Journal of Geophysical Research: Planets*, vol. 122, pp. 789–793, Apr. 2017. DOI: 10.1002/2016JE005216.

[2] E. C. Stern et al., "Ablation and heating during atmospheric entry and its effect on airburst risk," NASA / NTRS, Tech. Rep. ARC-E-DAA-TN42804, May 2017, Presented at the IAA Planetary Defense Conference, Tokyo, May 15–19, 2017. [Online]. Available: https://ntrs.nasa.gov/api/citations/20180002971/downloads/20180002971.pdf.

[3] M. Aftosmis, "Simulation-based height of burst map for asteroid airburst damage prediction," NASA / NTRS, Tech. Rep. 20170011263, n.d. Accessed: 2025-10-16. [Online]. Available: https://ntrs.nasa.gov/api/citations/20170011263/downloads/20170011263.pdf.

[4] P. Chodas, *Protecting earth from neo impacts*, https://www.kiss.caltech.edu/workshops/neo/presentations/ShortCourse_Chodas-Protecting_Earth.pdf, Accessed: 2025-10-16, n.d.

[5] European Space Agency (ESA), *Planetary defence — esa space safety*, https://esoc.esa.int/space-safety-planetary-defence, Accessed: 2025-10-16, n.d.

[6] NASA Planetary Defense Strategy & Action Plan Working Group, "Nasa planetary defense strategy & action plan," National Aeronautics and Space Administration, Tech. Rep. NA, May 2023, Accessed: 2025-10-16. [Online]. Available: https://www.nasa.gov/wp-content/uploads/2023/06/nasa_-_planetary_defense_strategy_-_final-508.pdf.

[7] K. Kaiho and N. Oshima, "Site of asteroid impact changed the history of life on earth: The low probability of mass extinction," *Scientific Reports*, vol. 7, no. 14855, 2017. DOI: 10.1038/s41598-017-14199-x. [Online]. Available: https://www.nature.com/articles/s41598-017-14199-x.

[8] C. Parsons, *A moment that changed earth*, https://www.nsf.gov/science-matters/moment-changed-earth, Accessed: 2025-10-16, National Science Foundation, Jun. 2022.

[9] Lunar and Planetary Institute (LPI), *Barringer (meteor) crater — lpi*, https://www.lpi.usra.edu/science/kring/epo_web/impact_cratering/enviropages/Barringer/barringerstartpage.html, Accessed: 2025-10-16, n.d.

[10] Meteor Crater Barringer Space Museum, *Meteor crater — info*, https://meteorcrater.com/info/, Accessed: 2025-10-14, 2025.

[11] Encyclopædia Britannica, Inc., *Tunguska event*, https://www.britannica.com/event/Tunguska-event, Accessed: 2025-10-16, n.d.

[12] NASA, *Five years after the chelyabinsk meteor, nasa leads efforts in planetary defense*, NASA Solar System Exploration, https://www.nasa.gov/solar-system/five-years-after-the-chelyabinsk-meteor-nasa-leads-efforts-in-planetary-defense/, Accessed: 2025-10-14, 2018.

[13] T. Phillips, *Nasa tracks russian meteor plume*, https://science.nasa.gov/science-research/earth-science/15aug_russianmeteorplume/, Accessed: 2025-10-16, NASA, Aug. 2015.

[14] A. S. Rivkin and et al., "Planetary defense with the double asteroid redirection test (dart) mission and prospects," *Scientific Reports*, vol. 13, 2023. DOI: 10.1038/s41598-023-47252-5. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC9978022/.

[15] NYC Department of City Planning, *Population — nyc planning*, https://www.nyc.gov/content/planning/pages/planning/population, Accessed: 2025-10-16, n.d.

[16] A. Wellerstein, *Nukemap*, https://nuclearsecrecy.com/nukemap/, Accessed: 2025-10-14, 2025.

[17] M. Paine, *The frequency and consequences of cosmic impacts since the demise of the dinosaurs*, Accessed: 2025-10-16, n.d.

[18] P. Chodas, *Protecting earth: An overview of near-earth object risks and mitigation*, https://www.kiss.caltech.edu/workshops/neo/presentations/ShortCourse_Chodas-Protecting_Earth.pdf, Caltech KISS Workshop on Near Earth Objects Short Course, 2023.

[19] M. Hajdukova, V. Sterken, P. Wiegert, and L. Kornos, "The challenge of identifying interstellar meteors," *Planetary and Space Science*, 2020. DOI: 10.1016/j.pss.2020.105126. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0032063320302737.

[20] NASA, *Nasa confirms dart mission impact changed asteroid's motion in space*, https://www.nasa.gov/news-release/nasa-confirms-dart-mission-impact-changed-asteroids-motion-in-space/, Oct. 2022.

[21] NASA, *Planetary defense – dart*, https://science.nasa.gov/planetary-defense-dart/, Feb. 2025.

[22] D. Humphries, *Double asteroid redirection test (dart)*, https://www.ebsco.com/research-starters/earth-and-atmospheric-sciences/double-asteroid-redirection-test-dart, 2024.

[23] My NASA Data, *Albedo values*, https://mynasadata.larc.nasa.gov/basic-page/albedo-values, Accessed: 2025-10-14, 2019.

[24] The Nine Planets Project, *M-type asteroids: Facts, information, history & definition*, https://nineplanets.org/m-type-asteroids/, Accessed: 2025-10-14, 2020.

[25] NASA, *Chapter 1: The solar system - nasa science*, https://science.nasa.gov/learn/basics-of-space-flight/chapter1-3/, Accessed: 2025-09-15, 2025.

[26] The Nine Planets Project, *S-type asteroids: Facts, information, history & definition*, https://nineplanets.org/s-type-asteroids/, Accessed: 2025-10-14, 2021.

[27] K. Howells, *What are asteroids made of?* https://www.planetary.org/articles/what-are-asteroids-made-of, The Planetary Society, Accessed: 2025-09-15, n.d.

[28] The Nine Planets Project, *C-type asteroids: Facts, information, history & definition*, https://nineplanets.org/c-type-asteroids/, Accessed: 2025-10-14, 2020.

[29] NASA, *Neo basics*, https://cneos.jpl.nasa.gov/about/neo_groups.html, Accessed: 2025-09-14, n.d.

[30] I. P. Williams, "The velocity of meteoroids: A historical review," *Atmospheric Chemistry and Physics*, vol. 4, pp. 471–494, 2004. [Online]. Available: https://acp.copernicus.org/articles/4/471/2004/acp-4-471-2004.pdf.

[31] A. P. Kartashova, E. M. Khodachek, D. G. Zverev, and A. S. Nikitin, "Study of injuries from the chelyabinsk airburst event," *Science of the Total Environment*, vol. 621, pp. 891–898, 2018. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S0032063317303884.

[32] National Research Council, *Defending Planet Earth: Near-Earth Object Surveys and Hazard Mitigation Strategies*. Washington, D.C.: The National Academies Press, 2010. [Online]. Available: https://nap.nationalacademies.org/read/12842/chapter/50.

[33] The Planetary Society, *Nasa's planetary defense budget growth*, https://www.planetary.org/articles/nasas-planetary-defense-budget-growth, Accessed: 2025-10-14, 2023.

[34] NASA, *Nasa's neowise celebrates 10 years, plans end of mission*, NASA Missions, https://www.nasa.gov/missions/neowise/nasas-neowise-celebrates-10-years-plans-end-of-mission/, Accessed: 2025-10-14, 2023.

[35] Jet Propulsion Laboratory, *Asteroid radar research: Introduction*, NASA JPL, https://echo.jpl.nasa.gov/introduction.html, Accessed: 2025-10-14, 2023.

[36] NASA, *Neo surveyor*, NASA Science Missions, https://science.nasa.gov/mission/neo-surveyor/, Accessed: 2025-10-14, 2024.

[37] R. L. Jones, S. R. Chesley, T. Grav, et al., *Synergies between the vera c. rubin observatory, roman space telescope, and neo surveyor for planetary defense*, arXiv preprint arXiv:2508.14412, 2025. [Online]. Available: https://arxiv.org/pdf/2508.14412.

[38] Phys.org, *Rubin observatory to unleash millions of solar system objects in vivid detail*, Phys.org, https://phys.org/news/2025-06-rubin-observatory-millions-solar-vivid.html, Accessed: 2025-10-14, 2025.

[39] Jet Propulsion Laboratory, *Twenty years of tracking near-earth objects*, NASA JPL, https://www.jpl.nasa.gov/news/twenty-years-of-tracking-near-earth-objects, Accessed: 2025-10-14, 2018.

[40] Associated Press, *Newly discovered asteroid once thought to pose threat will not hit earth, nasa says*, AP News, https://apnews.com/article/newly-discovered-asteroid-nasa-space-6f81d82acc55c899764d2a900d33f1d0, Accessed: 2025-10-14, 2025.

[41] K. Howley, M. Bruck Syal, et al., *Scientists design conceptual asteroid deflector and evaluate it against massive potential threat*, https://www.llnl.gov/article/44186/scientists-design-conceptual-asteroid-deflector-and-evaluate-it-against-massive-potential-threat, Accessed: 2025-10-16, Lawrence Livermore National Laboratory, Mar. 2018.

[42] J. Howe, ""space and defense"," *Space & Defense Journal*, no. Article 1120, n.d. Accessed: 2025-10-16.

[43] Johns Hopkins University Applied Physics Laboratory, *Impactor spacecraft — dart mission*, https://dart.jhuapl.edu/Mission/Impactor-Spacecraft.php, Accessed: 2025-10-16, Johns Hopkins APL, n.d.

[44] The Planetary Society, *Dart: Nasa's test to stop an asteroid from hitting earth*, https://www.planetary.org/space-missions/dart, Accessed: 2025-10-16, The Planetary Society, n.d.

[45] R. De Gaudenzi, "Satellite networks: Past, present and future challenges – part 2: Broadband systems," *IEEE Aerospace and Electronic Systems Magazine*, 2025. DOI: 10.1109/MAES.2025.3565491.

Grand Challenge Title

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